

AUTONOMOUS RC CAR Meeting 6 PRESENTATION

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OUR GOAL AS A TEAM

Create an autonomous vehicle that can navigate through an unstructured environment.

Overall Steps

Step 01

Can it run autonomously without using the remote, with sensors and a microcontroller board to which we can upload code?

Step 02

Can it detect and identify obstacles correctly?

Step 03

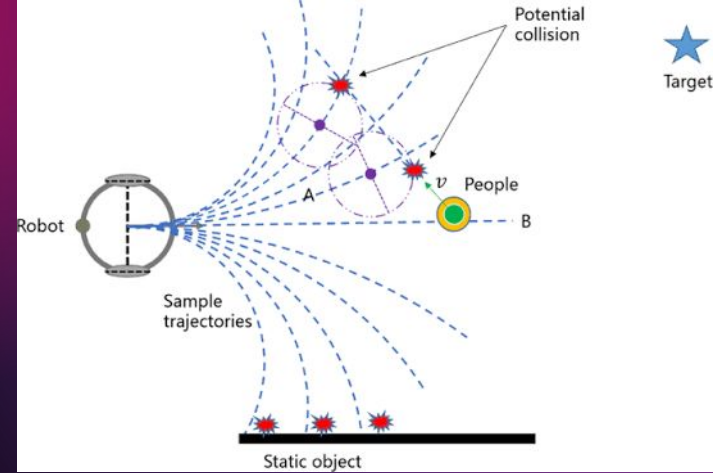
Can it apply that obstacle detection to develop a software that can then avoid the obstacles?

Step 04 (Bonus)

Can it create paths based on given targets?

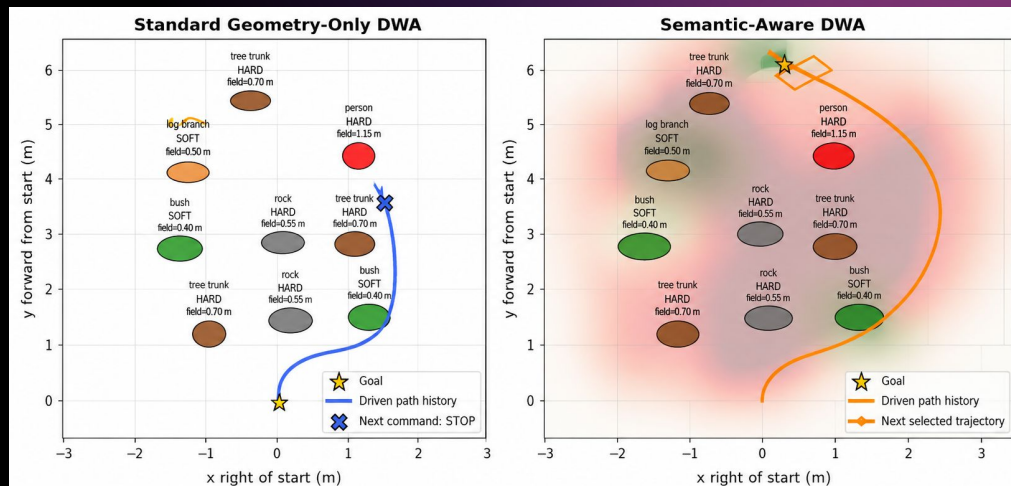
DWA Path Planning Algorithm

- Dynamic Window Approach:
 - Local path planning algorithm
 - Predicts short term trajectories
 - Selects safest route toward target
- Limitations in unstructured environments:
 - Short sighted
 - Runs on interval
 - Predicts best path for next interval of travel
 - Gets to next point and repeats until goal is reached
 - Chooses shortest path
 - Can commit to poor routes in dense environments



Modified DWA Path Planning Algorithm

- Modified Dynamic Window Approach:
 - Detect objects using trained YOLO model
 - Estimate object locations with LiDAR
 - Build a semantic map of environment
 - Score possible paths using semantic information
 - Select safest path



Next Steps:

- Keep training the AI obstacle detection so that it does a better job with avoiding trees and other unstructured obstacles.
- Write code and finish hardware so we can implement the LIDAR sensor into the design. Use it for better and more efficient obstacle detection.

THANK YOU

Any questions?